When Does QoS-Aware Scheduling Earn Its Complexity? A Two-Tier Scheduler for Private 5G Factory Networks

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Abstract—Private 5G networks for factories carry traffic whose classes want qualitatively different things: guaranteed-bitrate uplink video, motion control with millisecond deadlines, and elastic best-effort. Proportional fair (PF), the de facto default in LTE and NR and the scheduler shipped by OpenAirInterface, has no notion of a rate contract or a deadline. A large literature proposes QoS-aware alternatives, but little of it asks the engineering question that decides whether to deploy one: in which regimes does QoS-awareness change outcomes a user feels, and in which is it overhead? We design a two-tier scheduler—a network-utility-maximisation program setting per-flow rate targets on a ∼1 s horizon, and a per-slot drift-plus-penalty tracker that realises them under a symbol-accurate PRB and PDCCH budget—and evaluate it against RR, PF and a class-aware gradient baseline. Our headline result is that the value of QoS-awareness is a hump, not a slope: in deep overload and under light load the two-tier design is indistinguishable from PF on contract satisfaction, and it decides outcomes only in a moderate-overload band, where it honours 10/10 GBR contracts against PF's 5/10. We further show that the soft GBR penalty used by utility-based schedulers reduces the strategic program to a fractional knapsack whose optimum abandons the lowest-spectral-efficiency flow outright, that no reweighting of that penalty can prevent this, and that a max-min satisfaction stage does— raising the worst-served flow from 0% to 40% of its contract. We report two negative results and release the simulator and all scripts.

Index Terms—private 5G, MAC scheduling, quality of service, network utility maximisation, Lyapunov optimisation, configured grants, OpenAirInterface

I. Introduction

A private 5G network on a factory floor carries a workload that public mobile broadband does not. Mobile robots stream machine-vision and LIDAR uplink at a sustained rate; motion-control loops exchange small payloads under packet delay budgets (PDB) of a few milliseconds; firmware pushes and log uploads soak up whatever is left. These classes do not simply want more throughput—they want different things. A best-effort flow wants its share of the residue. A guaranteed-bitrate (GBR) flow wants a specific rate or nothing of value. A delay-critical flow wants its bytes before a deadline, after which they are worthless.

The scheduler that ships in OpenAirInterface (OAI), and the de facto default across LTE and NR, is proportional fair (PF) [1]. PF ranks users by instantaneous rate over smoothed average rate; its fixed point maximises $\sum_i \log R_i$ [2]. It is a good answer to a question the factory is not asking, because it has no representation of a rate contract or a deadline.

A substantial literature adds QoS-awareness by modifying the PF metric with an urgency term—M-LWDF [3], EXP/PF and the Exp-rule [4], and the variants surveyed in [5]. A parallel thread casts allocation as network utility maximisation (NUM) [2] or as Lyapunov drift-plus-penalty control [6]–[8]. What is scarcer is an answer to the question an operator actually faces before adopting any of it: a two-tier scheduler means an LP solver, a control loop, and a non-trivial port into a real-time MAC. When is that worth paying for?

This paper answers that question for a private-5G factory target, and in doing so surfaces a structural pathology in a mechanism that utility-based schedulers commonly rely on.

Contributions

1) A two-tier scheduler (Section IV) composing a NUM program on a ∼1 s horizon with a per-slot drift-plus-penalty tracker, granting resources per UE with a MAC logical-channel multiplexer so that grant granularity and DCI cost match the 5G MAC. The strategic tier is dependency-isolated for an OAI port.
2) A regime characterisation (Section VI-A): the value of QoS-awareness is a hump, not a slope. At the shipped load (deep overload) and at one third of it (light load), the two-tier design and PF are indistinguishable on contract satisfaction. Only in a moderate band does the scheduler decide outcomes—10/10 GBR contracts against PF's 5/10. A study reporting mean throughput alone would have concluded the design was not worth building.

3) A structural result on soft GBR penalties (Section IV-C). In the standard formulation—maximise utility, minus a large penalty on GBR shortfall—the penalty outweighs the utility by a factor we measure at 3.3×10^7. The program is then effectively minimise total shortfall bits under a PRB budget: a fractional knapsack whose optimum is greedy in spectral efficiency and therefore a vertex. The lowest-SE GBR flow is not degraded but abandoned, at exactly 0—where the same flow receives 23% under the utility term alone. We show no reweighting of the penalty can remove this, and that a max-min satisfaction stage does, lifting the worst-served flow from 0% to 40% of contract at a cost of 4% of cell throughput.
4) Evidence that Configured Grants are structurally decisive (Section VI-B). When the control channel binds, the two-tier design meets 30/30 deadlines against PF's 2/30. The mechanism is a double bypass—zero PDCCH and no buffer-status-report round-trip—and PF-class schedulers have no equivalent for either.
5) Two negative results, reported because they cost us months: an adaptive dual-ascent GBR penalty and a spectral-efficiency tilt were both implemented, measured, and rejected (Section VI-F).
6) An open, reproducible artifact: the simulator, the scheduler library, and a script per result table.

II. Background and Related Work

A. Channel-opportunistic scheduling

Round Robin is channel-blind and wastes capacity on faded users; Max-C/I maximises cell throughput and starves the cell edge; PF [1] interpolates via $r_{\text{inst}}/R_{\text{avg}}$ and is the practical default. All three share one limitation: none can be told "this flow needs 8 Mbps" or "this packet dies in 10 ms."

B. QoS-aware single-tier schedulers

M-LWDF [3] multiplies the PF metric by head-of-line delay and a weight; the Exp-rule [4] uses an exponential urgency term. These handle delay well. For rate contracts they share a subtler defect: a multiplicative urgency metric of the form $(r_{\text{inst}}/R_{\text{avg}}) \cdot (1 + w \cdot \text{deficit})$ has an equilibrium that equalises a weighted $1/R_{\text{avg}}$ across flows. It cannot drive delivered rate to an arbitrary, unequal target vector however large w is. This saturation argument is why our Tier-2 is a drift-plus-penalty integrator rather than a tuned PF variant; we include a class-aware "Gradient" baseline of exactly this shape to make the difference measurable.

C. Optimisation-based and hierarchical RRM

NUM [2] states allocation declaratively as $\max \sum_i U_i(r_i)$ under capacity constraints, but yields a rate vector, whereas a wireless scheduler must decide which UE to serve this slot over a stochastic channel. Drift-plus-penalty control [6] and greedy primal-dual scheduling [8] address the per-slot problem but need a target to track. The natural composition—slow declarative program feeding a fast tracker—is the pattern behind NVS [9] and the O-RAN non-RT/near-RT RIC split [10]. Our contribution is not the pattern but its concrete instantiation against a symbol-accurate 5G NR resource model, and the empirical characterisation of when it pays.

D. The OAI baseline

OAI's [11] NR MAC schedules with a PF metric over per-UE grants. Our ProportionalFair baseline is modelled on it, and all schedulers we compare grant per UE with one DCI each, so the comparison isolates QoS-awareness rather than grant granularity.

III. System Model

We consider a single NR cell [12] with a static TDD pattern. Time is slotted; each slot offers a set of physical resource blocks (PRBs) in one direction, with the special slot contributing partial DL and UL symbol counts. Let $d(i) \in \{\text{DL}, \text{UL}\}$ be flow i's direction and C_d the capacity of direction d in PRB-symbols per second over the TDD cycle. Per-UE SNR evolves as an AR(1) process in dB and maps through a link-adaptation staircase [13] to SE_i, bits per PRB-symbol, discounted by target BLER.

Each flow is uni-directional and carries a 5G QoS profile [14]: a class $c(i) \in \{\text{GBR}, \text{Delay}, \text{PF}\}$, a guaranteed flow bit rate G_i (GBR only), a packet delay budget PDB_i, a 5QI-style priority_level, and an optional slice identifier. D_i denotes offered demand.

Two control-plane budgets are modelled explicitly because both bind in practice. First, a per-slot PDCCH/CCE budget: every dynamic grant costs one DCI at an aggregation level that grows as SNR falls. Second, the uplink buffer-status-report round-trip: a real gNB learns of UL data only via a delayed BSR, so the scheduler sees a lagged view of UL backlog. Semi-persistent scheduling (SPS, "Configured Grants" [15], [16]) bypasses both—a CG user transmits on a standing allocation with no DCI and no BSR. Section VI-B shows this double bypass is decisive.

IV. Two-Tier Scheduler Design

A. Tier-1: the strategic solve

Tier-1 answers, every $\sim$1 s, what rate should each flow target? Decision variables are target rates $r_i \geq 0$ (bps) with shortfall slacks $s_i, s_\sigma \geq 0$ for GBR and slice floors.

All solves share one feasible set, parameterised by a per-flow floor:

$$\begin{aligned}
\mathcal{F}(\text{floor}): \quad & \textstyle\sum_{i:d(i)=d} r_i/\mathrm{SE}_i \le C_d && \forall d && (1)\\
& r_i \le D_i && \forall i && (2)\\
& r_i + s_i \ge G_i && \forall i\colon c(i)=\text{GBR} && (3)\\
& r_i \ge \text{floor}_i && \forall i\colon c(i)=\text{GBR} && (4)\\
& \textstyle\sum_{i\in\sigma} r_i/\mathrm{SE}_i + s_\sigma \ge \phi_\sigma C_d && \forall \sigma && (5)
\end{aligned}$$

Constraint (1) is where symbol-accurate accounting enters: r_i/SE_i is the resource a target rate consumes, and C_d counts special-slot symbols. Slice floors (5) are capped at the slice's own demand and kept soft, so an idle slice holds nothing and a busy slice borrows freely—the allocation remains work-conserving.

B. Shortfall before utility: an ordering, not a weight

The conventional formulation is a single objective,

$$\max \;\textstyle\sum_i w_{c(i)} \log(r_i + \epsilon) \;-\; \sum_i p_i s_i, \qquad (6)$$

with p_i "large" so that GBR floors are effectively hard whenever feasible. On our factory workload we measure the two terms at 718 and 2.4×10^{10} respectively: a ratio of 3.3×10^7. That is not a weighting. It is a lexicographic ordering expressed as a magnitude, and it has two consequences.

The first is numerical. Posed as (6), the program is badly conditioned: on a three-flow instance small enough to solve in closed form, two interior-point solvers both returned optimal_inaccurate, disagreed with each other by 3.6× on one flow's rate, and under-served the highest-weighted (Delay) class by 28% against the analytic optimum. Rescaling does not fix this—a ratio between objective terms is a property of the model, not of the units. Stating the order directly does. We solve, over $\mathcal{F}$,

$$\text{B1:}\quad S^\star = \min \;\textstyle\sum_i p_i s_i + p_\sigma \sum_\sigma \mathrm{SE}_\sigma s_\sigma \qquad (7)$$

$$\text{B2:}\quad \max \;\textstyle\sum_i w_{c(i)} \log(r_i + \epsilon) \;\text{ s.t. } (7) \le S^\star \qquad (8)$$

after which both solvers agree to within 100 bps of the analytic optimum. A useful side effect: p_i's absolute magnitude becomes irrelevant and only the ratios across flows matter. Both slacks are converted to bits per second before being weighed—a slice slack is natively in PRB-symbols—so that p_i and p_σ are quoted in one currency. Compared raw, the crossover between them sits at $p_i \cdot \mathrm{SE}_i$, which would make slice-versus-GBR priority a function of the radio channel rather than of policy.

C. Soft GBR floors are a knapsack

The second consequence is structural, and it is the more interesting one. Once the penalty dominates, (7) is $\min \sum_i (G_i - r_i)$ subject to $\sum_i r_i/\mathrm{SE}_i \le C$: a fractional knapsack. Its optimum is the greedy one—buy rate where it is cheapest per PRB, i.e. in descending spectral efficiency—and therefore a vertex of the feasible polytope. Flows are served in full or abandoned outright, with at most one partially served flow per SE tier. Table I shows the solved targets on our factory workload: a fully served head, one boundary tier, and a zeroed tail.

TABLE I
Tier-1 targets under (6) on factory_robots at the shipped load. The allocation is an exact spectral-efficiency staircase—note it orders by SE, not SNR: ue2 (18 dB) and ue4 (16 dB) share an MCS step and receive identical targets.

SE	Flows (SNR)	Target
48.6	ue1, ue5 (22, 24 dB)	100%
37.8	ue3, ue6, ue8, ue10 (19–21 dB)	100%
29.7	ue2, ue4, ue9 (16–18 dB)	50–67%
21.6	ue7 (14 dB)	0%

Two controls confirm the mechanism. First, sweeping p over six decades (1 to 10^6) leaves the targets identical: the solution is fixed by the knapsack structure, not by the penalty magnitude, so no choice of p is a tuning opportunity. Second, we drop the shortfall term entirely and maximise the utility alone. That allocation is not generous to the cell edge—the stationary condition of $\sum \log r_i$ under a capacity constraint is $r_i \propto \mathrm{SE}_i$, so the utility is itself SE-favouring, and the worst-channel flow receives 23% of its contract against the best flow's 89%. But it serves that flow something. Adding the shortfall term takes the same flow to exactly 0.

The distinction matters for attribution. Cell-edge starvation under utility-based scheduling is often ascribed to "weighted-log pathology"; our measurements locate it instead in the shortfall term, which is what converts a merely SE-favouring allocation into a vertex one that abandons outright.

It also explains why penalty-shaped fixes cannot work. Since the optimum is a vertex, reweighting p selects a different vertex and nothing more (Section VI-F). Removing an abandonment requires changing the feasible region—a constraint, not a weight.

D. A max-min GBR floor

We therefore prepend a max-min satisfaction stage. Let $\hat{G}_i = \min(G_i, D_i)$ be flow i's reachable contract; the demand cap matters, or an under-offered GBR flow pins the level at its own unreachable ratio and drags the set down. Stage A solves

$$t^\star = \max t \quad \text{s.t.} \quad \begin{aligned} & r \in \mathcal{F}(0), \;\; t \in [0,1], \\ & r_i \ge t\hat{G}_i \;\; \forall i\colon c(i)=\text{GBR} \end{aligned} \qquad (9)$$

and sets $\text{floor}_i = \alpha\, t^\star \hat{G}_i$ in (4). $t^\star$ is the largest fraction of contract that every GBR flow can hold simultaneously; $t^\star = 1$ means the GBR set is jointly feasible.

This is enabled by default, and the reason it is safe is that it self-disables: whenever $t^\star = 1$ the floor binds nothing and the result is identical to omitting the stage. It costs something only in genuine GBR overload. α trades the guarantee against throughput, with $\alpha = 0$ recovering the single-stage behaviour exactly.

All programs are posed in normalised units—rates as multiples of the largest contract, capacity usage as a fraction of each direction's budget—so every coefficient is $O(1)$. This is not cosmetic: posed in raw bps, a unit-scale t against 10^7-scale rates made the solver report optimal on a $t^\star$ that was non-monotone in capacity, which is impossible for a max-min level.

E. Tier-2: drift-plus-penalty rate tracking

Each flow carries a virtual queue Q_i, a control accumulator in bits rather than a buffer of data. Per slot of duration Δ: grow $Q_i += r_i\Delta$; clamp $Q_i \leftarrow \min(Q_i, \mathrm{ceil}_i)$; serve; drain $Q_i \leftarrow \max(0, Q_i - \mathrm{delivered}_i)$. Because Q_i is a non-saturating integrator it can enforce an arbitrary target vector, which is precisely what the multiplicative metrics of Section II cannot do.

The clamp is subtle. Q_i is bounded by what the flow legitimately should have delivered over the trailing Tier-1 window W but did not: $\mathrm{ceil}_i = \max(0, \min(r_i W, A_i^W) - D_i^W)$ with A^W, D^W bits arrived and delivered in the window. Using a windowed arrival count rather than instantaneous backlog is essential: a bursty video flow's buffer momentarily empties between frames, and clamping to that near-zero backlog erases its legitimate rate-tracking debt, letting continuous flows starve bursty GBR ones. This was a real regression in our implementation.

F. Per-UE grants and the MAC multiplexer

The 5G MAC grants per UE—one DCI, one transport block—so Tier-2 does too. For Delay flows a head-of-line urgency bonus enters the queue before ranking, $\tilde{Q}_i = Q_i + w_\mathrm{d}(\mathrm{HoL}_i/\mathrm{PDB}_i)^\kappa \max_j Q_j$, scaled by the system-wide maximum queue rather than the flow's own small target—otherwise a low-rate control flow could never out-compete bulk. UEs are then ranked by $M_u = \left(\sum_{i\in u}\tilde{Q}_i\right)\cdot \mathrm{SE}_u$, which is simultaneously opportunistic and rate-tracking, and served greedily subject to PRB and CCE budgets. A granted UE's transport block is filled across its flows by logical-channel prioritisation [15]: by priority_level, then by $\tilde{Q}_i$. The baselines use the same multiplexer (with plain backlog as the tiebreak, since they carry no virtual queues), so all schedulers are strictly comparable on DCI cost.

G. Configured grants

Periodic flows receive standing allocations sized to their contracted floor, granted in priority tiers with proportional scale-back so that no flow is dropped for being late in an enumeration order. A viability floor governs the mechanism: if a tier's reservations would be scaled below a threshold, the tier runs dynamically instead—unless dropping it would overrun the CCE budget, in which case SPS's zero-DCI property makes it the lesser evil. SPS thus self-gates, engaging where it helps and standing aside where fixed reservations would starve the dynamic pool.

V. Evaluation Methodology

A. Simulator and fidelity discipline

We evaluate in a purpose-built discrete-event simulator whose job is comparative questions, not absolute ones. The design principle is to model, to good accuracy, every resource the scheduler competes for, and nothing else. Modelled: the slot-by-slot PRB grid including the TDD special slot; the per-slot PDCCH/CCE budget with variable DCI aggregation level; the UL BSR round-trip as a per-flow delay plus Bernoulli loss, bypassed by configured grants; CQI report delay and loss, with BLER computed from the mismatch between the MCS the scheduler picked and the true SNR at transmission; per-UE AR(1) SNR through an MCS staircase; per-flow buffers with head-of-line timestamps and PDB expiry. Not modelled: LDPC and I/Q, HARQ as a full state machine (a fixed delay plus BLER discount instead), per-packet RLC segmentation, MU-MIMO, mobility and inter-cell interference.

The discipline cuts both ways, and the PDCCH budget is the case in point: modelling it is what reveals configured grants to be decisive (Section VI-B), and omitting it would have led to the opposite conclusion. The same held for the BSR round-trip, which was our largest fidelity gap until we closed it; omitting it understates SPS.

B. Scenarios

Three workloads isolate three bottlenecks. factory_robots: 10 mobile robots, 24 flows—uplink camera and LIDAR (GBR), downlink motion-control (Delay), best-effort uploads and a TCP flow—with an SNR spread of 14–24 dB and uplink demand roughly twice carrier capacity as shipped. sensor_dense: 30 small periodic uplink sensors with a 15 ms PDB, sized so the control channel binds before the data channel. latency_bound: 8 interactive 5 Mbps streams with a 12 ms PDB sharing a saturated downlink with 80 Mbps of bulk.

The load sweep varies offered load around the shipped operating point. Because capacity in a real deployment is fixed by spectrum, we report the axis as load relative to the shipped point, where higher is worse. Uplink runs with an 8-slot ($\approx$4 ms) BSR round-trip and an 8-slot CQI report delay throughout.

C. Metrics

We report contract-oriented metrics, and this is a deliberate methodological choice: mean delivery ratio hid every finding in this study. A GBR flow's contract is its GFBR, counted as met at $\geq 95\%$; a Delay flow's is $\geq 99\%$ on-time delivery within PDB. We report the count of flows meeting their contract, the minimum delivery across the GBR set, and p99 head-of-line delay. Mean and total throughput appear as context, not as the verdict—a

TABLE II
Load sweep on factory_robots. Load is relative to the shipped operating point; higher is worse. "met" counts GBR flows delivering $\geq 95\%$ of GFBR. −mm pins the max-min stage off.

Load	Scheduler	Total	met	mean	min
1.00×	PF	69.3 M	0/10	37%	0%
	TwoTier	66.7 M	0/10	44%	40%
	TwoTier −mm	74.2 M	0/10	53%	0%
0.67×	PF	93.9 M	4/10	55%	4%
	TwoTier	94.1 M	0/10	65%	60%
	TwoTier −mm	95.8 M	0/10	68%	43%
0.50×	PF	111.2 M	5/10	72%	31%
	TwoTier	120.1 M	10/10	82%	78%
0.33×	PF	124.7 M	10/10	85%	83%
	TwoTier	125.4 M	10/10	85%	83%

TABLE III
sensor_dense: 30 periodic uplink sensors, 15 ms PDB. On-time requires $\geq 99\%$ delivery within PDB.

Scheduler	Total	on-time	min deliv.	p99
RoundRobin	6.9 M	0/30	71%	15.0 ms
PF	9.2 M	2/30	85%	15.0 ms
TwoTier	9.6 M	30/30	100%	5.0 ms

TABLE IV
latency_bound: 8 interactive streams (12 ms PDB) versus 80 Mbps of bulk on a saturated downlink.

Scheduler	on-time	mean	p99	bulk DL
RoundRobin	3/8	84%	12.0 ms	22.8 M
PF	5/8	86%	12.0 ms	24.6 M
TwoTier	8/8	100%	10.5 ms	15.2 M

scheduler can post a healthy 86% mean while the missing 14% is one starved safety-critical flow.

VI. Results

A. The value of QoS-awareness is a hump

Table II sweeps offered load. The shape is the paper's headline. At the shipped load, GBR demand far exceeds capacity and no scheduler honours a contract; at one third of it everyone has slack and all converge to 10/10. The scheduler decides outcomes only in between: at 0.50× load the two-tier design honours 10/10 contracts against PF's 5/10, with a minimum delivery of 78% against 31% and +8.9 Mbps of cell throughput.

The 0.67× row is where the metric becomes the discussion. The two-tier design meets fewer knife-edge contracts than PF (0/10 against 4/10) while delivering a strictly better distribution: minimum 60% against 4%, mean 65% against 55%. A 95% threshold rewards a scheduler that abandons some flows so others clear the bar. For a factory where every robot matters, "all robots degraded gracefully" beats "four robots perfect, six offline"—so the contract count is necessary but not sufficient, and must be read alongside the minimum.

Engineering implication. Since capacity is fixed by spectrum, the operator's lever is admission control: keep peak offered load in the moderate band, because that is where the scheduler pays for itself. A cell that systematically runs at $\gtrsim 2.5\times$ its capacity has a capacity-planning problem that no MAC fixes.

B. Control-channel-limited: a capability PF lacks

Table III reports sensor_dense, where the DCI/CCE budget binds before the data channel. The two-tier design meets 30/30 deadlines with a 5 ms p99 tail; PF meets 2/30 and pins at the 15 ms PDB. This is not a tuning difference. Each periodic flow receives a standing allocation that costs zero PDCCH per slot and needs no BSR round-trip; PF is throttled by both budgets at once and has no mechanism to bypass either.

We regard this as the strongest practical result in the paper: for any deployment with dense periodic small-payload traffic—sensors, PLCs, AGV telemetry—configured-grant support is not an optimisation but a prerequisite, and capacity planning that counts only PRBs will mis-size the cell.

C. Deadline-blindness is silent

Table IV reports latency_bound. The two-tier design meets 8/8 deadlines; PF meets 5/8. The trade is explicit and deliberate: bulk throughput falls from 24.6 Mbps to 15.2 Mbps to fund the interactive set.

The dangerous property is that PF's failure is quiet. Its control-flow mean delivery is 86%—a number that reads as healthy on a dashboard—but the missing 14% are aged-out packets, and in a teleoperation or motion-control loop those are precisely the safety-relevant ones. Mean delivery is not a safety metric.

D. Cell-edge starvation and the max-min floor

Table V gives the per-flow picture at the shipped load and makes the knapsack result of Section IV-C concrete. Without the max-min stage the two lowest-SE flows land at exactly 0%—not degraded, abandoned—while high-SNR peers run at 71–80%. With the stage enabled every GBR flow lands between 50% and 56%: a 6-point spread across a 10 dB SNR range, against PF's 0–91%.

The cost is stated plainly: at the shipped load PF carries more total throughput than the two-tier design (69.3 vs 66.7 Mbps). The max-min floor gives up 4% of aggregate capacity to hold the worst-served flow at 40% of contract rather than 0%. Whether that is the right trade is a deployment question—a robot whose video is switched off entirely is a failure in a way that a uniformly degraded fleet is not—but it is a trade, and we do not present it as a free win.

TABLE V
Per-flow delivery as a fraction of GFBR, factory_robots at the shipped load, worst channel first. −mm pins the max-min stage off.

Flow	SNR	RR	PF	TwoTier −mm	TwoTier
ue7	14 dB	23%	28%	0%	53%
ue4	16 dB	51%	62%	0%	50%
ue9	17 dB	3%	3%	55%	50%
ue2	18 dB	57%	68%	53%	54%
ue6	19 dB	37%	46%	50%	55%
ue3	20 dB	63%	74%	69%	53%
ue10	20 dB	89%	0%	69%	53%
ue8	21 dB	3%	3%	69%	53%
ue1	22 dB	77%	91%	71%	56%
ue5	24 dB	56%	67%	80%	56%
mean		46%	44%	53%	53%

Two further effects are visible. The mixed-flow UEs (ue8–ue10, carrying a GBR flow and a best-effort flow) collapse to 0–3% under the QoS-blind baselines, because their MAC multiplexer fills the transport block by raw backlog and a continuously backlogged best-effort flow wins every time. And the bill for the max-min floor is paid by the high-SNR flows: ue1 drops from 91% under PF to 56%.

E. Sensitivity to control-plane fidelity

Because our BSR and CQI models are approximations, we sweep them. BSR delay matters and is close to linear: over 0–16 slots PF's minimum delivery falls 66%→30% and its contract count breaks 8/10→5/10, while the two-tier design moves 80%→77% and 10/10→8/10 and loses about 1% of throughput against PF's 10%. SPS-served flows are invariant to BSR by construction. BSR loss is near-inert on this workload (0–20% loss moves nothing) because continuously backlogged video keeps buffers non-empty, so a lost report rarely changes the eligibility bit.

CQI staleness is likewise small here, and the reason is a property of the channel rather than of the scheduler: our AR(1) model produces per-slot SNR innovations well below the ∼3 dB spacing of MCS thresholds, so even a stale report usually picks the right MCS. A conservative semi-persistent MCS for SPS—a mobility hedge—is neutral at 0 dB margin and actively harmful beyond 1 dB, because larger reservations cross the SPS viability floor. We report this as a negative result for our deployment class: the hedge is worth paying for under mobility, and a static factory is exactly where it is not.

F. Two negative results

We implemented and rejected two mechanisms, both aimed at cell-edge starvation. An adaptive dual-ascent penalty escalates p_i on whichever flow is actually missing. It raises minimum delivery (0%→34%) but meets 0/10 contracts in both overloaded rows, because dual ascent drives toward equal normalised shortfall while a GBR contract is a step function—equalising shortfall parks every flow just below its floor so none clears it. A spectral-efficiency tilt $p_i \propto \mathrm{SE}_i^k$ with $k < 0$ rescues the cell edge only by relocating starvation to the next-worst tier.

Section IV-C explains why neither could have worked: both reweight a linear penalty whose optimum is a vertex, so they select a different victim rather than removing the abandonment. We think this is the transferable lesson—in an infeasible allocation problem, a weight chooses whom to sacrifice; only a constraint prevents sacrifice—and it applies to any scheduler enforcing rate contracts through penalty terms.

VII. OAI Implementation Results

[Placeholder. The simulation results of Section VI motivated an OpenAirInterface implementation; its measurements are reported here.]

A. Implementation

[Stub.] What the port required: where Tier-1 runs relative to the MAC scheduler thread and how its targets are published; how Tier-2 attaches to the gNB's per-slot scheduling path; how configured grants are set up and released; and which of the simulator's abstractions — symbol-accurate capacity, the CCE budget, the delayed BSR view — map onto real gNB state rather than having to be re-derived. Worth stating explicitly which parts of the scheduler/ library lifted unchanged and which did not, since dependency isolation was an explicit design goal.

B. Experimental Setup

[Stub.] Testbed: gNB hardware and radio unit, OAI release, carrier and TDD configuration, UE population and type, and how the factory workload of Section V was reproduced on real UEs. Any deviation from the simulated scenarios should be named — particularly offered load, SNR spread, and whether the deep-overload operating point is reachable at all.

C. Results

[Stub.] The same contract-oriented metrics as Section VI: GBR contracts met, minimum delivery across the GBR set, on-time fraction and p99 head-of-line delay, measured against the stock OAI PF scheduler on identical traffic. Tier-1 solve time and its margin against the re-solve cadence belong here too, since real-time feasibility is a claim simulation cannot make.

D. Discussion

[Stub.] The load-bearing question is which simulated conclusions survive contact with hardware: whether the moderate-overload band remains the regime where the design pays (Section VI-A); whether configured grants retain their structural advantage once real DCI and BSR timing apply (Section VI-B); and whether the max-min floor behaves as Section VI-D predicts on a measured rather than modelled SNR spread. Divergence is more interesting than agreement, and should be attributed to a specific unmodelled effect.

VIII. Threats to Validity

Simulation, not silicon. Results are comparative and architecture-level. They establish which scheduler wins and roughly by how much; they do not predict absolute throughput or microsecond-accurate latency, and they do not validate real-time feasibility or UE interop. An OAI integration is the necessary next step.

Warm-up and sampling. A single run reads GBR delivery about 5 points below steady state, with ± 2 point per-window scatter (channel coherence is comparable to the run length). We verified over 60 consecutive windows that the scheduler-versus-scheduler gap is stable, so comparative claims hold; absolute percentages should be read as soft.

Link model. Spectral efficiency is a fitted staircase rather than a 3GPP TBS table extract—adequate for comparative work, insufficient for absolute claims.

Workload realism. Three synthetic scenarios isolate three bottlenecks cleanly, but trace-driven factory workloads would strengthen external validity. Our video model uses a fixed I-frame multiplier without GOP structure.

Unmodelled control-plane latency. UL k_2 grant-to-transmission timing is not modelled; including it would widen the configured-grant advantage further, so its omission is conservative with respect to our conclusions.

A contract-dimensioning caveat. Even at three times capacity, about 15% of GBR video bytes expire on PDB. We verified this is scheduler-independent—a lone video flow drops identically under RR, PF and our design—because an I-frame burst is 3–4$\times$ the GFBR, making a "GFBR plus tight PDB" contract internally inconsistent. This is an admission and contract-dimensioning problem, not a scheduling one, and it bounds what any scheduler can achieve on such a workload.

IX. Conclusion

We set out to decide whether a QoS-aware two-tier scheduler is worth building for a private-5G factory, or whether the proportional-fair scheduler that OpenAirInterface ships today is good enough. The answer is conditional, and we have characterised the conditions.

QoS-awareness is not a uniform improvement. In deep overload and under light load it is indistinguishable from PF on contract satisfaction, and a study reporting mean throughput alone would have concluded it was not worth the complexity. Contract-oriented metrics reveal three regimes where it changes outcomes sharply: control-channel-limited deployments, where configured grants deliver 30/30 deadlines against PF's 2/30 via a capability PF structurally lacks; latency-bound mixed deployments, where PF misses deadlines silently; and moderate-overload GBR deployments, where the strategic tier honours contracts the cell could otherwise have met. The load-bearing features are configured grants and a deadline-aware Tier-2—not, as we initially assumed, the sophistication of the strategic program.

Our analytical contribution is a diagnosis with a fix: the soft GBR penalty that utility-based schedulers rely on turns the strategic program into a fractional knapsack whose optimum abandons the worst-channel flow entirely, no reweighting of that penalty can prevent it, and a max-min satisfaction stage can—at a stated cost in aggregate throughput. We expect this to generalise to any scheduler that enforces rate contracts through penalty terms.

Future work is the OpenAirInterface integration the strategic tier was isolated for, trace-driven workloads, and an admission-control gate: our max-min level $t^\star$ is a ready-made infeasibility detector, since $t^\star < 1$ quantifies how far a GBR set is from satisfiable before any flow has been starved to reveal it.

Artifact Availability

The simulator, the scheduler library and all scenario configurations are available at https://github.com/artpark-hub/5g-qos-stack. Every table in Section VI is regenerated by a single command: Tables II, III and IV by scheduler_study.py; Table V by maxmin_study.py and compare_schedulers.py; Table I and the penalty sweep of Section IV-C by maxmin_study.py; the sensitivity results of Section VI-E by bsr_study.py and cqi_study.py. A dated engineering log recording every finding, regression and rejected design in this paper is included in the repository.

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